



Basic Emergency Care

Pediatric emergencies

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Learning objectives

- Recognize cardiac arrest in children
- Perform high-quality pediatric CPR
- Manage airway in infants and children
- Understand compression–ventilation ratios
- Know when to activate emergency response
- Define foreign body inhalation
- Recognize partial vs complete airway obstruction
- Perform age-appropriate emergency management

CPR

- **Pediatric Basic Life Support (PBLS)** is a rescue procedure which has purpose of preventing the anoxic brain damage by promoting the return of spontaneous circulation and breathing in cases of cardiac arrest.
- Protocols differentiated according to age:
 - baby: from 0 to 28 days
 - infant: from 1 month to 12 months
 - youth: from 12 months to puberty (about 10–11 years)

CPR

- **A:** Airway
- **B:** Breathing
- **C:** Circulation

CPR

- 1- Determine responsiveness: By calling the child, if unresponsive try a rigorous painful stimuli
- 2-Position the child in a supine position (careful with movement)
- 3-Check airway: If compromised airway and/or no breathing, Immediately attempt to open the airway with basic airway maneuvers
- 4-Check breathing: Look, feel, listen
- 5-Check Pulse: Carotid or Brachial pulse

All should take no more than half a minute!

CPR



CPR

Recognizing cardiac arrest in children:

- Unresponsive
- No breathing or gasping
- No pulse (or HR <60 in infants)

Immediately start **CPR** and call for **HELP**

CPR

Chest compressions:

- 30:2 ratio if 1 rescuer (30 compressions per 2 breaths). 15:2 if 2 rescuers.
- 100-120 compressions per minute (2 per second)
- Push down at least $\frac{1}{3}$ of AP diameter of chest (Infant 4cm, child 5cm)
- Allow complete chest recoil after each compression to allow cardiac filling
- Minimize interruptions

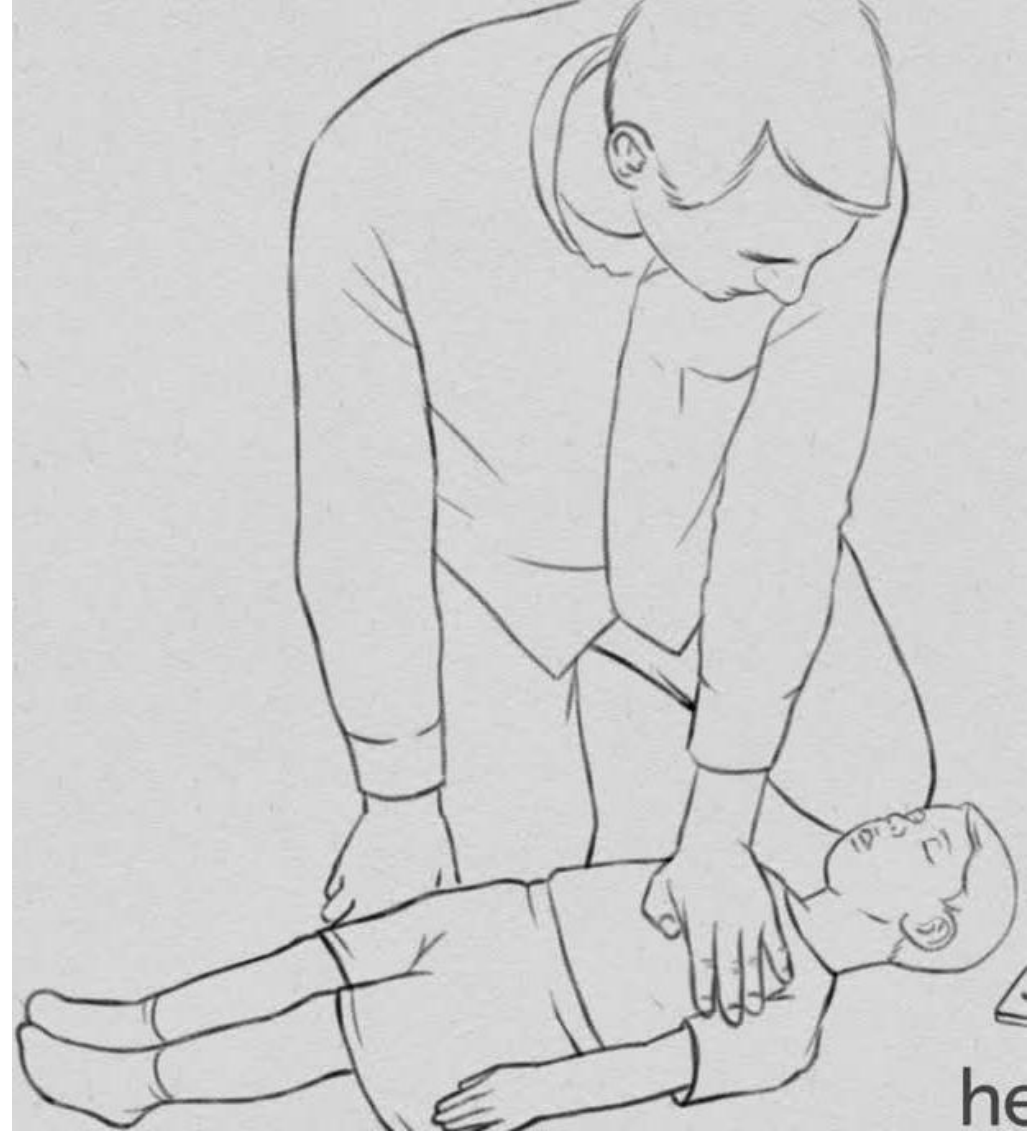
CPR



2 Finger



2 Thumb encircling

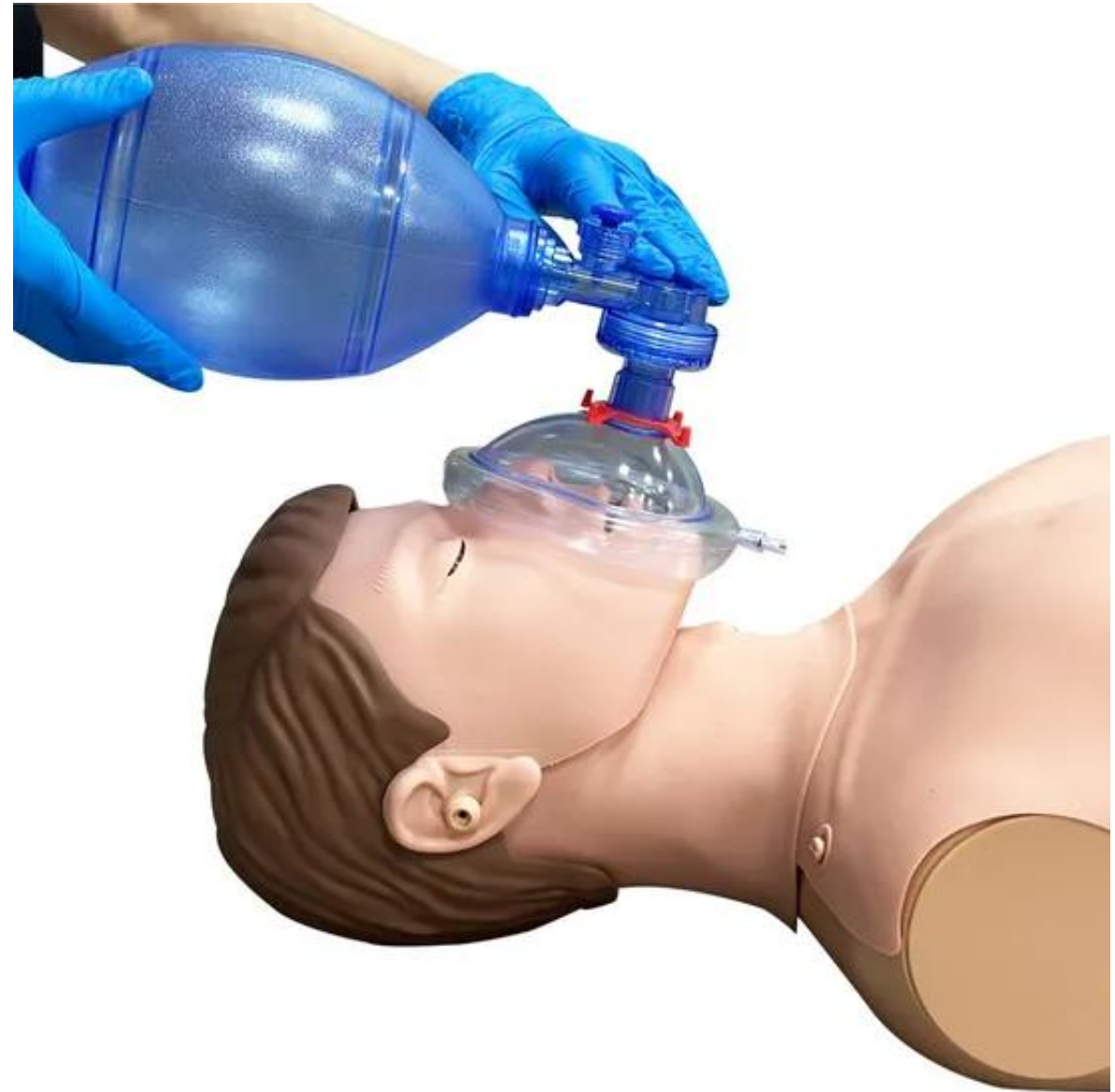


CPR

Breaths:

- The breaths should be delivered slowly, Slow breaths will allow delivery of an
- adequate volume of air and ensure effective lung and chest expansion.
- Use bag and mask ventilation if available

CPR



CPR

- **If pulse present but no breathing:**
- 1 breath every 2–3 seconds
- Recheck pulse every 2 minutes

CPR

- Check rhythm with DC machine
- Give pharmacological treatment when appropriate



CPR

- **When to Stop CPR?**
- Return of spontaneous circulation (ROSC)
- Advanced team takes over
- You are exhausted
- Scene becomes unsafe

Foreign body inhalation

- **Foreign body inhalation (Aspiration)**
- Entry of a solid object into the airway causing partial or complete obstruction.
- Most common in 1-3 years of age.

Foreign body inhalation

Foreign body lodges in:

- Larynx → complete obstruction (life-threatening)
- Trachea
- Right main bronchus (more common)

Leads to:

- Air trapping
- Atelectasis
- Hypoxia
- Cardiac arrest (if severe)

Foreign body inhalation

Classic history:

- Child was eating or playing
- Sudden coughing episode
- Choking
- Respiratory distress

Foreign body inhalation

Signs of Airway Obstruction:

Partial Obstruction:

- Effective coughing
- Wheezing
- Stridor
- Crying or speaking possible

Encourage coughing

Foreign body inhalation

- **Complete Obstruction:**

- Cannot speak or cry
- Silent cough
- Cyanosis
- Clutching neck
- Rapid deterioration

Foreign body inhalation

Conscious infant:

Steps:

- Support head and neck
- 5 back blows
- Turn infant over
- 5 chest thrusts
- Repeat until object expelled

No abdominal thrust in infants!

Foreign body inhalation



5 back blows



5 chest thrusts

Foreign body inhalation

Conscious child:

Abdominal Thrusts (Heimlich Maneuver)

- Stand behind child
- Fist above umbilicus
- Quick inward & upward thrusts
- Repeat until object expelled.

Foreign body inhalation



Foreign body inhalation

- Heimlich maneuver:
- 1-Stand behind the child with arms directly under the child's axilla
- 2-Place the thumb against the victim's abdomen in the midline above the umbilicus and below the xiphoid process.
- 3-Grasp the fist with the other hand and exert a series of quick
- upward thrusts.
- Each thrust should be a separate distinct movement to relieve the obstruction.

Foreign body inhalation

If unconscious:

- Start **CPR** immediately

Key takeaways

- ✓ Most pediatric arrests are respiratory
 - ✓ Start CPR immediately if HR <60 with poor perfusion
 - ✓ Compression depth = 1/3 chest
 - ✓ 15:2 if 2 rescuers
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- ✓ Sudden coughing while eating = suspect foreign body
 - ✓ Encourage coughing if partial obstruction
 - ✓ Back blows + chest thrusts for infants
 - ✓ Heimlich maneuver for children
 - ✓ Start CPR if unconscious

Thank you